27 February 2026

The Editor
Classical and Quantum Gravity

Dear Editor,

I am submitting the enclosed Letter, *Gravity Field Dynamics and Bimetric Gravity: Dual Tetrahedral Field Chambers as the Key to the General Parameters*, for consideration in Classical and Quantum Gravity. The paper addresses a specific open problem in the Hassan–Rosen bimetric framework: the theoretical determination of the five free interaction coefficients β_n, which have been treated as continuous phenomenological parameters since the framework was established in 2012.

The paper introduces Gravity Field Dynamics (GFD), which derives gravitational coupling laws from the combinatorial geometry of the local tetrad frame. The central object is a coupling polynomial $f(k, s) = 1 + ks + k^2$, evaluated at $k = 4$ tetrahedral faces. Its static skeleton at the symmetric point yields the locked integer coefficients $\{\beta_n\} = \{1, 0, 16, 0, 1\}$, uniquely determining the ghost-free Hassan–Rosen bimetric potential. Its dynamic body, under a conformal map onto the acceleration variable, independently produces a GFD constitutive law $\mu(x) = x/(x + 1)$, a kinetic functional $F(y)$, and a zero-parameter covariant scalar-tensor action derived entirely from the same polynomial. The UV bimetric potential and the IR scalar-tensor action are both projections of this single geometric object.

A distinctive feature of this work is that the tetrad geometry is made explicitly visible. The paper writes the tetrad one-forms as four half-space inequalities ($v_1 = x + y + z$, $v_2 = x – y – z$, $v_3 = –x + y – z$, $v_4 = –x – y + z$) and evaluates them at four million sampled points to produce a point-cloud reconstruction of the dual tetrahedral field chamber. No mesh or curve fitting is involved. The $4 \times 4 = 16$ geometric coupling channels that determine $\beta_2 = 16$ are visible in the figure before the algebraic proof begins. An interactive three-dimensional reconstruction is provided as supplementary material, alongside a Python validation script that independently verifies every algebraic claim in the manuscript.

The result extends beyond the bimetric parameter lock. The $N = 2$ bimetric completion produces a distinct, falsifiable prediction at the transition scale: $\mu(1) = 1/\phi^2 \approx 0.382$, compared to $\mu(1) = 1/2$ for the $N = 1$ screened limit. This testable distinction between the single-metric and bimetric sectors arises directly from the GFD capacity function and requires no additional assumptions.

The derivation is self-contained and algebraic, suited to the Letter format. I believe the paper will be of interest to CQG's readership in bimetric gravity, massive gravity, and modified gravity phenomenology. I confirm that this work has not been published elsewhere and is not under consideration by another journal.

Sincerely,

Stephen Nelson
stephen@fielddynamics.org
ORCID: 0009-0008-6190-3269